

CO5 – AT3

Course Code: MLA0201

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Course Name: Fundamentals of Machine Learning

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Mock- Interview

Q1. How would you use Bayesian Networks to predict machine failures?

I would represent machine conditions such as temperature, vibration, pressure, and maintenance history as nodes in a Bayesian Network. These variables would be connected to a Machine Failure node using conditional probabilities. When sensor values are observed, Bayesian inference calculates the probability of failure. This helps industries perform predictive maintenance before a major breakdown occurs.

Q2. How do Markov Random Fields help in image denoising?

A Markov Random Field represents each **pixel as a node** connected to its neighboring pixels. Nearby pixels usually have similar intensity values, so the model uses these neighborhood dependencies to identify noisy pixels. By considering neighboring pixel values, the system updates abnormal values and produces a smoother image while preserving important structures.

Q3. How would you apply an HMM in speech recognition?

In speech recognition, the actual spoken sounds can be treated as **hidden states**, while audio features are the observations. The HMM uses **transition probabilities** to model how speech sounds change over time and **emission probabilities** to connect hidden sounds with observed audio signals. Algorithms such as the **Viterbi algorithm** can then determine the most likely sequence of spoken states.

Q4. How would you explain CRFs for part-of-speech tagging?

Conditional Random Fields are used for **sequence labeling**, where each word receives a label such as noun, verb, or adjective. A CRF considers the features of the current word and the relationship between neighboring labels. This allows the model to use contextual information and produce more accurate labels for an entire sentence.

Q5. How does conditional independence simplify probabilistic inference?

Conditional independence reduces the number of probabilities that must be calculated. For example, in a Bayesian Network:

Rain → Wet Road ← Sprinkler

Once the state of the Wet Road is considered appropriately, the model represents dependencies using local relationships instead of calculating every possible joint probability directly. This makes probabilistic inference more efficient and easier to model.

Q6. Compare directed and undirected graphical models in healthcare diagnosis. A

Bayesian Network uses directed edges and can represent relationships such as:

Disease → Symptoms

It is useful when cause-and-effect relationships are known.

A **Markov Random Field** uses undirected edges and represents mutual relationships between symptoms.

It is useful when variables strongly influence each other without a clear direction.

For medical diagnosis, Bayesian Networks are generally more suitable because diseases can naturally be modeled as causes of symptoms.

Q7. How can Bayesian inference handle missing patient data?

Bayesian inference can calculate probabilities using the available evidence without requiring every variable to be present. Missing symptoms are treated as **unknown variables**, and the model marginalizes over their possible values. Therefore, the system can still estimate the probability of a disease based on the observed symptoms and prior knowledge.

Q8. How would you use an HMM for customer purchase sequences?

Customer behavior can be represented using hidden states such as **Browsing, Interested, and Purchasing**. Observations may include product views, clicks, cart additions, and purchases. The HMM learns transitions between customer states and predicts the customer's likely next behavior. This information can improve product recommendations.

Q9. Why do CRFs often outperform HMMs in Named Entity Recognition?

HMMs model the probability of observations and hidden states and usually require stronger assumptions about the data. CRFs directly model the conditional probability of labels given the input features. CRFs can use multiple contextual features such as surrounding words, capitalization, and word patterns. Therefore, they often achieve better accuracy in Named Entity Recognition.

Q10. How can graphical model learning and generalization improve fraud detection?

Graphical model learning identifies relationships between variables such as **transaction amount, location, device, and previous fraud history**. The learned dependencies help estimate the probability of fraud for new transactions. Generalization techniques such as validation and regularization help prevent overfitting, allowing the fraud detection system to perform reliably on previously unseen transactions.
